Nursing Accelerated Renal training questions

1. Which one of the following makes the glomerular capillaries effective in filtration?

a. The glomerular capillary hydrostatic pressure is less than 32 mmHg.
b. The capillaries have tight endothelial junctions.
c. The capillaries have fenestrated endothelial junctions.
d. The plasma protein osmotic pressure is more than 60 mmHg.

2. All the following are functions of the kidney EXCEPT:

a. Removal of waste products and foreign chemicals from the blood.
b. Control of the concentration of most constituents of body fluids.
c. Regulation of arterial blood pressure.
d. Regulation of WBCs formation through secretion of erythropoietin.
e. Regulation of water and electrolyte balance.
f. Regulation of acid base balance.

3. The proximal tubule of the kidney nephron:

a. is impermeable to water and solutes.
b. is the site of maximum reabsorption of water (65%).
c. responds to ADH by reabsorbing a large volume of water.
d. responds to aldosterone by reabsorbing sodium and excreting potassium and hydrogen.

4. Movement of water from the tubular fluid through the descending loop of Henle into the interstitial fluid depends upon:

a. activity of the Na/K/2Cl co-transporter.
b. activity of the Na/Cl co-transporter.
c. movement of glucose through the tubular epithelial cell.
d. Passive diffusion only.
e. effect of ADH on the loop of Henle.

5. The basic theory of the nephron function depends on which of the following?

a. Filtration.
b. Reabsorption.
c. Secretion.
d. A, B, and C.

6. Which of the following defines the term glomerular filtration?

a. It is the movement of fluid and solutes from the bowman's capsule into the glomerular capillaries.
b. It is the movement of fluid and solutes from the distal tubules to the blood.
c. It is the movement of fluid and solutes from the glomerular capillaries into the bowman's capsule.
d. It is the movement of fluid and solutes from the proximal tubules to blood.

7. Match the following renal structures to its scientific name:

A. Juxtaglomerular apparatus
B. Cortical nephron
C. Renal corpuscle
D. Juxta medullary nephron
E. Nephron

1. Nephron that has long loop of Henle reaching the tip of renal papilla.
2. Combination of the glomerulus and the Bowman's capsule.
3. Structural and functional unit of the kidney.
4. Nephron that lies totally in the cortex of the kidney.
5. Macula densa of early distal tubules, afferent arteriole, efferent arteriole.

8. All the following are correct concerning glomerular filtrate EXCEPT:

a. Glomerular filtrate is isotonic.
b. It is plasma minus proteins.
c. It contains water and all its solutes (wanted and unwanted).
d. It contains proteins.
e. It does not contain blood cells or platelets.

9. Which of the following is correct concerning glomerular filtration rate?

a. It equals 180 liters per minute.
b. It equals 180 ml per hour.
c. It is the amount of fluid filtered by all nephrons of both kidneys per minute.
d. It is the amount of fluid filtered by all nephrons of one kidney/ minute.

10. Glomerular filtration is directly proportional to -------------.

a. Net filtering pressure.
b. Glomerular capillary permeability.
c. Effective filtration surface area.
d. A, B, and C.

11. Which of the following is correct concerning the function of proximal tubules of the kidney?

a. Absorption of 65% of sodium, potassium, chloride, calcium and water.
b. Complete reabsorption of glucose and amino acids.
c. Absorption of 90% of bicarbonate and phosphate.
d. Secretion of organic acids like uric acid, creatinine and drugs as penicillin.
e. A, B, C, and D.

12. Which of the following is correct concerning the thin segment of loop of Henle?
   a. It is impermeable to water.
   b. It is permeable to water by passive transport only.
   c. It transports salt actively.
   d. They absorb 85% of water.

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14. Renal tubular secretion occurs to all the following EXCEPT:
   a. Drugs as penicillin.
   b. Hydrogen ion.
   c. Organic substances as urea, uric acid, and creatinine.
   d. Potassium ion.
   e. Proteins.

15. Which of the following is correct concerning renal distal tubules and cortical collecting ducts?
   a. Everything is absorbed under the control of hormones.
   b. Sodium is absorbed under the control of aldosterone.
   c. Water is absorbed under the control ADH.
   d. Hydrogen and potassium are secreted under the control of aldosterone.
   e. Calcium is absorbed under the control of parathyroid hormone.
   f. A, B, C, D, and E.

16. The reflex that initiates urination when the bladder fills is triggered by:
   a. Increased blood pressure in the bladder wall.
   b. Stretch receptors in the bladder wall.
   c. High urine osmolarity.
   d. Decreased detrusor muscle tone.

17. The "micturition center" that coordinates urination reflexes in the brainstem is located in the:
   a. Cerebellum.
   b. Thalamus.
   c. Pons.
   d. Medulla oblongata.

18. The spinal "micturition center" that coordinates urination reflexes is located in the:
   a. Cervical segments (C2-4).
   b. Sacral segments (S2-4).
   c. Lumbar segments (L2-4).

d. Thoracic segments (T2-4).

19. Which nerve fibers are primarily responsible for micturition by contracting the detrusor and relaxing the internal sphincter?

a. Pelvic nerves (parasympathetic).
b. Hypogastric nerves (sympathetic).
c. Pudendal nerve (somatic).
d. Splanchnic nerves.

20. The voluntary control of external sphincter of the urinary bladder is helped by ---------------------------.

a. Pelvic nerves (parasympathetic).
b. Hypogastric nerves (sympathetic).
c. Pudendal nerve (somatic).
d. Splanchnic nerves.